Enhancing Fetal ECG Extraction: Novel UNET Architecture using Continuous Wavelet Transforms performing dimentionality reduction

Tanish Shah
Department of Information Technology
Sardar Patel Institute of Technology
Mumbai, India
tanish.shah07@icloud.com

Pratik Pujari
Department of Computer Science
Sardar Patel Institute of Technology
Mumbai, India
pratikpujari1000@gmail.com

Atharv Raotole
Department of Information Technology
Sardar Patel Institute of Technology
Mumbai, India
atharva.r29@gmail.com

Diya Joshi
Department of Bioinformatics
DY Patil University
Navi Mumbai, India
diyapjoshi.27@gmail.com

Om Sheladia
Department of Electronics Engineering
Veermata Jijabai Technological Institute
Mumbai, India
omsheladia10@gmail.com

Devam Shah
Department of Mechatronics Engineering
MPSTME
Mumbai, India
devam.shah5740@gmail.com

***Abstract*—Pregnancy-related fetal heart monitoring is essential for the earliest identification of any possible cardiac abnormalities and the implementation of appropriate treatments to ensure a safe delivery. Extracting fetal electrocardiograms (ECGs) from maternal belly ECGs is a widely studied method for fetal cardiac surveillance. However, it is challenging because fetal signals can be overshadowed by maternal ECG signals and noise. We present a novel approach for efficient separation of an one-channel maternal abdomen ECG into maternal and fetal segments. This addresses the complexities of the extraction process. We implemented a U-NET architecture combined with continuous wavelet transform, resulting in a reduction in training parameters and enhancing the efficiency of the baseline model. We evaluate our suggested architecture's foetal ECG detection performance against that of other cutting-edge deep learning models. Notably, our model incorporates the detection of P waves, a feature missing in existing methods. Our findings show that the suggested model performs better than any other proposed architecture with respect to F1 scores, recall, and accuracy. This suggests that our approach is effective in extracting fetal ECGs from single-channel maternal abdominal ECGs. This advancement in fetal cardiac monitoring can contribute significantly to the early identification of potential issues, allowing for timely preventive measures and ensuring healthier births.**

***Keywords*—Electrocardiogram, fetal P and R-waves detection, U-Net, skip-connection, Continuous Wavelet Transform Introduction**

I. Introduction

In 2021, the world witnessed around 1.9 million stillbirths, which refers to the birth of a baby with no signs of life at 28 weeks or later [1]. Proper care could have prevented many of these stillbirths. According to the latest data, the global stillbirth rate in the previous year was 13.9 stillbirths per 1,000 total births [1]. This means one stillborn infant every 72 births, or one every 17 seconds [1]. However, this number could be lower than the actual stillbirths, as they are often underreported. Most of these stillbirths occur in the intrapartum period such as child birth. Therefore, fetal Heart rate needs to be monitored at all times in order to ensure that the fetus remains completely healthy during pregnancy.

Due to invasive methods causing pain and chorioamnionitis [2] in women, non- invasive methods is opted instead. In non-invasive method, electrodes are placed on the maternal abdomen to collect electrical activity of the fetal heart. Although at times the contact may break causing temporal irregularities during the recording. Fetal ECG extraction is essential as it can be used to reduce fetal mortality rates [3]. The underlying issue here is that extraction of Fetal ECG is slightly tricky as it is polluted with noise. One of the major contributor to this noise is the maternal electrocardiogram (mECG) and contraction from the uterus. Also, the abdominal signal incorporates the noise from sources such as electrohysterogram (EHG), baseline drift and also various muscular activities occurring within the mother's body.

Extraction Algorithms are used to filter out the maternal ECG from the Fetal Electrocardiogram (fECG) signal which can be difficult to perform if the fECG is covered by the mECG signal [4]. Our proposed work makes use of a U-Net model with skip connections between encoder and decoder layers which can recognize and extract the fECG signal. The 1-dimensional input is fed into the model onto the encoder and decoder layers. The model is also trained in such a way that it doesn't discard but learns the irregularities observed in the P waves and the T waves. The proposed model can effectively extract the fECG signal from the mECG and removes all the noise contained in the initial signal.

II. Background Information

A. Cardiotocography

Fetal electrocardiograms are a valuable tool for assessing the electrical activity of the fetal heart. They provide a complete assessment of the structure and function of the heart [5]. Fetal health state could be measured based on the fetal heart rate and uterine contractions which could provide valuable information about the fetus [6]. Ultrasonic fetal cardiotocography (CTG) was widely accepted as a non-invasive method to track fetal heart rate and uterine contractions in the 1960s [7]. Certain professionals noted that there was reduction of fetal death after the introduction of this method. Despite great expectations, this technique did not result in a rapid decrease in undiagnosed fetal hypoxia [8].

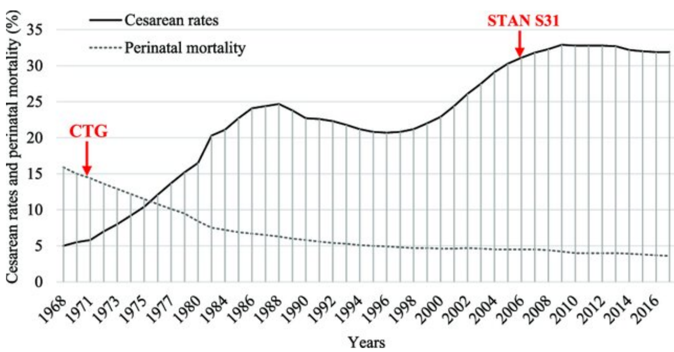

Fig. 1: Perinatal mortality and caesarean rates in the United States of America, based on [9]

Moreover, several studies showed that these FHR monitors were quite unreliable, and further it was difficult to interpret their output, which resulted in a dramatic increase in the number of Caesarian sections since the introduction of this technique (as indicated in Fig. 1).

B. Electrocardiogram

An alternative to ultrasonography is electronic foetal monitoring, which may offer more precise information about the foetal heart rate as well as more data about the electrical activity of the foetal heart, akin to an adult electrocardiogram (ECG). The three distinct waves that make up the ECG waveform are the P, QRS, and T-waves, which stand for atrial and ventricular contraction brought on by the heart's electrical activity travelling through it. The depolarization of the ventricles is represented by the QRS complex, which comes after the P-wave, which indicates the depolarization of the atria. The T-wave, on the other hand, indicates the repolarization of the ventricles. A typical ECG waveform is shown in Fig. 2

The interval between two successive R-R intervals provides a measure of the heart rate. It also includes certain morphological features derived from the ECG trace that give us more information about the underlying conduction system. These are listed in Table I.

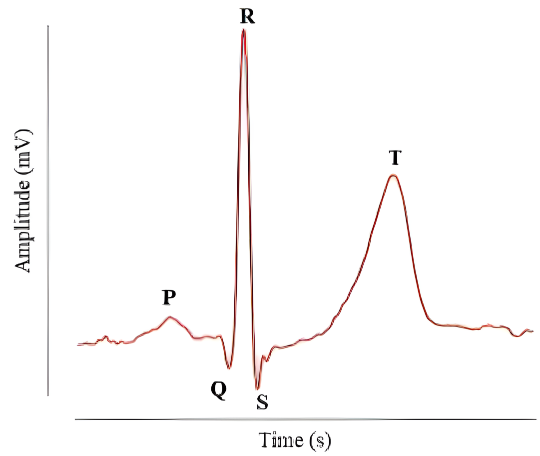

Fig. 2: A Typical ECG waveform (sampled from record NR05 NIFEADB)

Table I. ECG Waveform Characteristics

Name	Definition
P-wave height	Peak amplitude of P-wave from isoelectric line
P-wave duration	Duration between the onset and the end of the P-wave
PR interval	Duration between the onset of P-wave and onset of QRS complex
Q-wave height	Peak amplitude of Q-wave from isoelectric line
R-wave height	Peak amplitude of R-wave from isoelectric line
S-wave height	Peak amplitude of S-wave function from isoelectric line
QRS complex	Duration between the onset of the Q-wave and end the S-wave
T-wave height	Peak amplitude of T-wave from isoelectric line
T-wave duration	Duration between the onset and the end of the T-waves
ST segment	Duration between the end of the S-wave to the onset of the T-wave

Table II. Comparison of ECG Parameters: Fetal Heart vs. Adult Heart

Feature	Mean Value (Fetal heart)	Normal Range (Adult heart)
P-wave duration	43.9 ms	80-100 ms
PR interval	102.1 ms	120-200 ms
QRS complex	42.7 ms	60-100 ms
QT interval	224 ms	200-400 ms
T-wave duration	123.8 ms	100-250 ms

Table II shows the average values of heart rate for fetal ECG and maternal ECG. It also demonstrates the decomposition for each component of the ECG waveform. These characteristics of the adult heart have been thoroughly investigated, and normal value ranges for a healthy heart have been established. A deviation from these normal numbers could indicate a problem.

Monitoring and analyzing fetal ECGs aids in the detection of anomalies, reduces mortality, and ensures healthy births. It also helps in detecting early cardiac problems with the fetus and take appropriate measure to prevent unexpected consequences. Therefore, it is of utmost importance to monitor the fetal ECG during the period of pregnancy and at the time of delivery.

C. *Related Methods*

As depicted in Table III, fetal monitoring methods vary in applicability, with Phonocardiography (PCG) recording heart murmurs after 28-30 weeks, Cardiotocography (CTG) monitoring heart rate from 20 weeks, and Fetal Magneto cardiogram (FMCG) detecting magnetic fields post-20 weeks. Scalp ECG (SECG) is invasive and for labor only, while Non-invasive fetal ECG (NI-fECG) has low SNR, especially hindered by vernix caseosa at 28-37 weeks.

D. *Problems*

1) Abdominal Electrocardiogram : Electrodes placed on the mother's abdomen are used to collect the abdominal signals, as it is the closest non-invasive method to measure the fetal heart activity. However, the recorded signal contains not only the weak electrical activity of the fetal heart, but also the maternal electrical cardiac activity, and noise. The maternal component has a higher amplitude compared to the fetal component, and the frequency content of both components overlaps, making it difficult to separate them in time and frequency [10]. This overlap is visible in Figures 3 and 4.

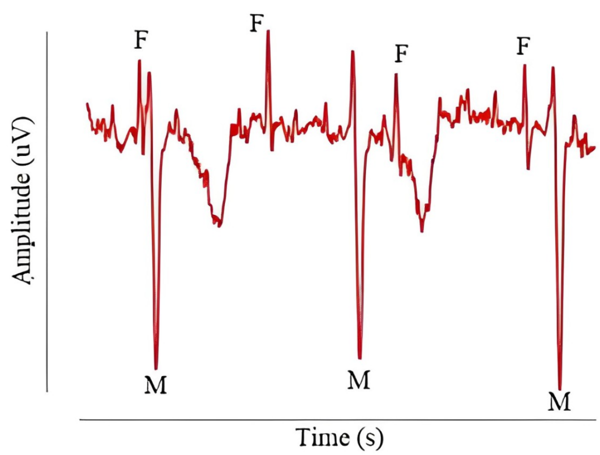

Fig. 3: The abdominal signal, taken from record r01 in ADFECGDB. Fetal QRS locations are marked as 'F', maternal QRS locations are marked as 'M'

2) Noise: In addition to the maternal and fetal ECG components, several sources of noise are also present in these recordings. Baseline wander (BW) is low-frequency noise (typically less than 1 Hz) commonly present in adult ECG recordings caused by respiratory movements. It also dominates fetal ECG recordings due to their low frequency, making it impossible to remove using a low pass filter. Abdominal recordings may be disturbed by stronger variations in the baseline due to electrical

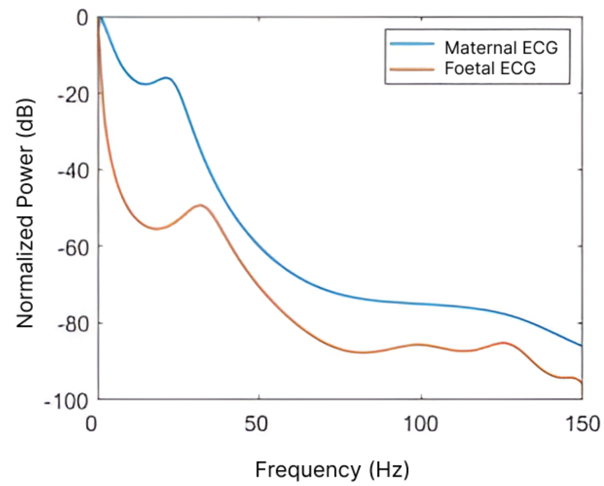

Fig. 4: A typical abdominal signal, and the frequency spectrum of the maternal ECG and the fetal ECG (taken from a scalp electrode).

activity arising from uterine contractions, known as electrohysterogram (EHG). To diagnose fECG, detecting EHG signals is necessary. EHG signals have low amplitudes, ranging from 0.1-5 Hz [11]. Directly filtering them out using a high-pass filter can cause loss of low-frequency features, which are important for accurate diagnosis. During labor, EHG signals undergo variations, making it difficult to detect fECG. The other major source of noise introduced into the signal is the 50/60 Hz (depending on the country) power line interference (PLI). This happens because of the interaction between the body and the ground as well as electromagnetic induction via recording wire loops. Harmonics of the fundamental 50/60 Hz frequency may also be present, which are usually generated by connected non-linear loads such as air conditioners [12], but in the case of NI-fECG, the spectral content above 100 Hz is almost negligible, and are hence usually filtered out using a low- pass filter instead. Muscle noise is more dominant in higher frequencies which are usually filtered out using a simple low-pass filter as part of pre-processing.

3) Electrode Placement: It is not yet fully understood what affects the quality of the abdominal ECG signal. However, studies have shown that there is a strong correlation between the signal-to-noise ratio (SNR) levels of the fetal ECG and the lead configuration used. The position of the fetus and the placenta also have an impact on the SNR levels. Anterior and fundal placental positions, as well as sideways and praevia/bipartia fetal positions, have been linked to weaker SNR [13]. There is currently no standard lead configuration that has been defined for NI-fECG recordings. However, it is widely recognized that the positioning of the electrodes plays a crucial role in obtaining good SNR recordings. Numerous placement schemes have been suggested in the literature and are available for review in reference [14].While some suggest the use of a fixed placement scheme for uniformity in recordings, others suggest that the position of electrodes needs to be varied in

Table III. Comparison of fetal monitoring methods

Method	Principle	Gestational age	Problems
Phonocardiography (PCG)	Sound recording of heart murmurs	28 – 30 weeks	• Lowest SNR of all techniques • Easily affected by acoustic noise sources • Fetal heart local required • Not routinely used for fetal health assessment
Cardiotocography (CTG)	Ultrasound	20 weeks	• Limited to heart rate • Safety not assessed • Not accurate, prone to errors
Fetal Magneto cardiogram (FMCG)	Detection of fetal heart's magnetic field using SQUID sensors	20 weeks	• Very expensive • Requires highly skilled personnel • Short term use only
Scalp ECG (SECG)	Scalp electrode placed on fetal scalp	Post membrane rupture (after water breaks)	• Invasive carries risk of infection and injury • Can only be used during labour • Single lead only
Non-invasive fetal ECG (NI-fECG)	Standard ECG electrodes placed on mother's abdomen	20 weeks	• Very low SNR • Hindered use between roughly 28th-37th week due to vernix caseosa

a case-by-case basis, depending on the position of the fetus [15].

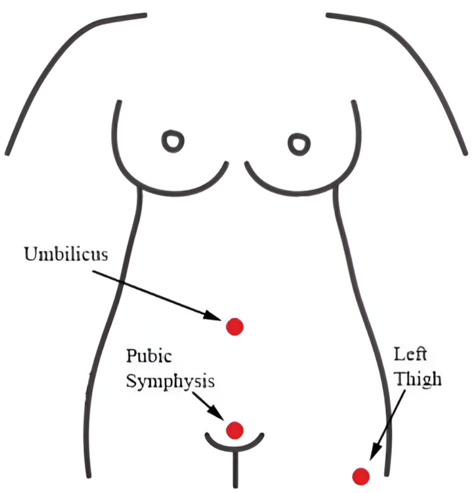

Fig. 5: Commonly used position for the placement of the common reference electrode

4) Extraction Algorithms: Monitoring the fetal requires signal processing as the fetal ECG contains noises and interferences having higher amplitude compared to the fECG signal. One majority of the noise incoming is from the maternal ECG (mECG) and other being uterine EGM during labor [16]. Hence the removal of mECG and high amplitude variations from abdominal signal is essential. This section reviews different extraction algorithms in order to remove the mECG interference.

Table. IV describes different extraction algorithms, its advantages and problems related to it. One of the filtering methods, Kalman Filter uses an estimation algorithm to make a series of measurements over time using joint distribution. On the backend it uses a Markov Model to maintain latent variables in state space. Adaptive Filtering has the ability to change the transfer function of the filter and common adaptive filters are recursive (RLS) and least mean square (LMS). Blind Source Separation works on the principle that signal is linear and a mixture of independent signal sources which in our case, the signals are mECG, fECG and noise. Wavelets are used to detect QRS waves and perform ECG classification and performs better than Fourier transform due to multi-resolution analysis. Lastly Machine learnings models have shown great results when compared to other classical filter. It has the ability to detect difficult peaks and irregularities in the dataset.

III. Methodology

A. Database

The Abdominal and Fetal ECG signals used in our proposed work is referred from a comparative analysis done by Medical University of Silesia, Poland [17]. The signals were recorded from five women during labor between 38 and 41 weeks of gestation. Each recording of 5-mins comprises of a direct fECG signal incoming from the fetal scalp and four signals received from maternal abdomen.

Fig. 6 shows a snippet of the recording clearly indicating one direct fECG and four abdominal signals. The abdominal electrode setup which consist of four electrodes located at navel, a common reference electrode and a reference electrode located near the pubic symphysis. All the recorded signals

Table IV. Comparison of Different Methods

Method	Key Points	Problems
Temporal	• Simple, low complexity	• Performance strongly depends on identification of mfECG • Templates may not account for varying morphology
Kalman Filtering	• Medium computational cost • Continuous estimation that depends only on identifying mQRS	• Highly dependent on model used • Several parameters to be optimized
Adative Filtering	• Once trained, it is very fast • Adaptive component allows it work well for variety of real signals	• Several parameters that affect stability and convergence that need to be optimized • Assumes linear relationship between chest and abdominal mECG
BSS	• Analysis based on higher-order statistics • Allows analysis in source domain	• Medium to high computational cost • Requires large number of channels • Can have unpredictable behaviour
Wavelets	• Allows multiresolution analysis • Low Computational complexity	• May deform the fECG which is not suitable for morphological analysis
Machine Learning	• Non-linear relationships can be established • Trained models are very positive	• Highly data-driven • Difficult to train

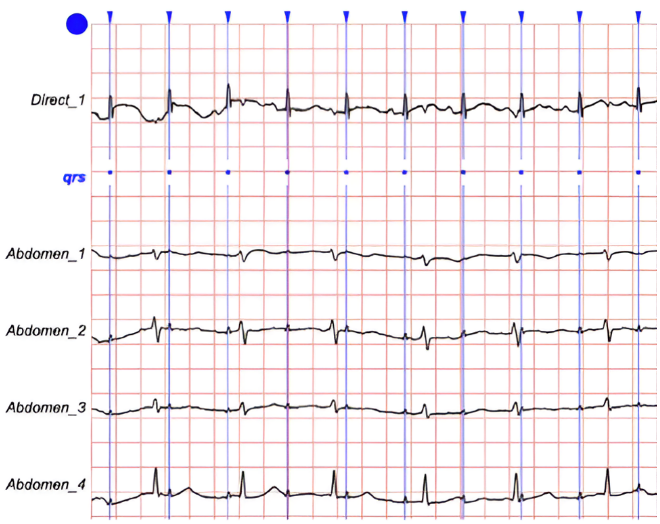

Fig. 6: Direct FECG and four abdominal recordings

were sampled at 1 KHz with 16-bit resolution. During the signal acquisition, the direct fetal electrocardiogram was acquired through a spiral electrode and its position was constant throughout all recordings. In contrast, the contact of the electrode was cut loose sometimes due to the movement of the patient [18]. The recorded fECG signals had some observed distortion due to this reason. At the time of labor, the fetal signal received was the strongest but overtime the abdominal signals showed some irregularities too due to uterine muscle contractions. The R-waves were automatically generated from the direct fECG signal through the on-line analysis method and verified by a group of cardiologists. We have sampled all the recording, observed the distortions and have meticulously chosen part of it's to ensure the recording has no high level of noise present in it. Sound to Noise ratio (SNR) was determined for each recordings and one with the highest value was chosen for further implementation.

B. Data Preprocessing

The signal data from 5 channels (1 Direct fetal ECG and 4 Abdominal ECG) is read and this data is then passed through a butterworth bandpass filter, which is applied to the data with a frequency range of 1 to 100 Hz and a sampling frequency of 1000 Hz. This process is repeated for different channels of the ECG data. The filtered data is then divided into sampled into size of 1024. These samples are then ready to be fed into the U-Net architecture for further processing and analysis. This approach ensures that the input to the U-Net architecture is clean, filtered, and appropriately segmented for optimal performance

C. Performance Assessment

The precision, F1-score, recall, and accuracy performance measures are evaluated across four distinct weight schemes in this proposed technique and other current suggested classifiers. In specifically, four notations are used to quantify the performance metrics: true-positive (t_p), true-negative (t_n), false-positive (f_p), and false-negative (f_n).

- **Precision**
  Precision is defined as the number of terms accurately identified positive to the total identified positively.

$$P_r = \frac{t_p}{t_p + f_p} \tag{1}$$

- **Recall**
The recall is defined as the percentage of accurately identified positive observations to the total observations in the dataset.
$$R_e = \frac{t_p}{t_p + f_n} \tag{2}$$

- **F1-score**
F1-score is defined as the average weight of recall and precision.
$$F_s = \frac{2 \cdot P_r \cdot P_e}{P_r + P_e} \tag{3}$$

D. Continous Wavelet Trasnform (CWT)

Incorporating Continuous Wavelet Transforms (CWTs) into the processing pipeline for ECG signal analysis can effectively reduce the number of training parameters required tor subsequent tasks such as classification or feature extraction- This reduction is achieved by capturing the relevant information in the time-frequency domain, thus reducing the dimensionality of the input data while preserving essential features. Let's consider a scenario where we have an ECG signal f (t) as our input data. Without incorporating CWTs. we might directly use the raw ECG signal for training a neural network or any other machine learning model. In this case. The number of training parameters would be determined by the length of the ECG signal and the sampling rate resulting in a potentially large number of parameters. However by incorporating CWTs. we can transform the raw ECG signal into its true-scale representation, effectively reducing the dimensionality of the input data. Mathematically the CWT of the ECG signal f (t) can be represented as:

$$CWT_f(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t)\psi^* \left(\frac{t-b}{a}\right) dt \tag{4}$$

Here, a represents the scale parameter controlling the width of the wavelet, b represents the translation parameter controlling the position of the wavelet, and $\psi^*(t)$ denotes the complex conjugate of the wavelet function $\psi(t)$. The result of this transformation is a time-scale representation of the ECG signal, where each point in the transformed data corresponds to a specific time and scale. By selecting appropriate scales and wavelet functions, we can focus on capturing the relevant frequency components of the ECG signal while discarding noise and irrelevant information.

This dimensionality reduction effectively reduces the number of training parameters required for subsequent tasks. Instead of using the raw ECG signal, which may contain redundant information, we can use the transformed time-scale representation obtained from the CWT as input to the model. This not only reduces computational complexity but also improves the model's ability to generalize by focusing on the most relevant features of the signal.

E. Architecture

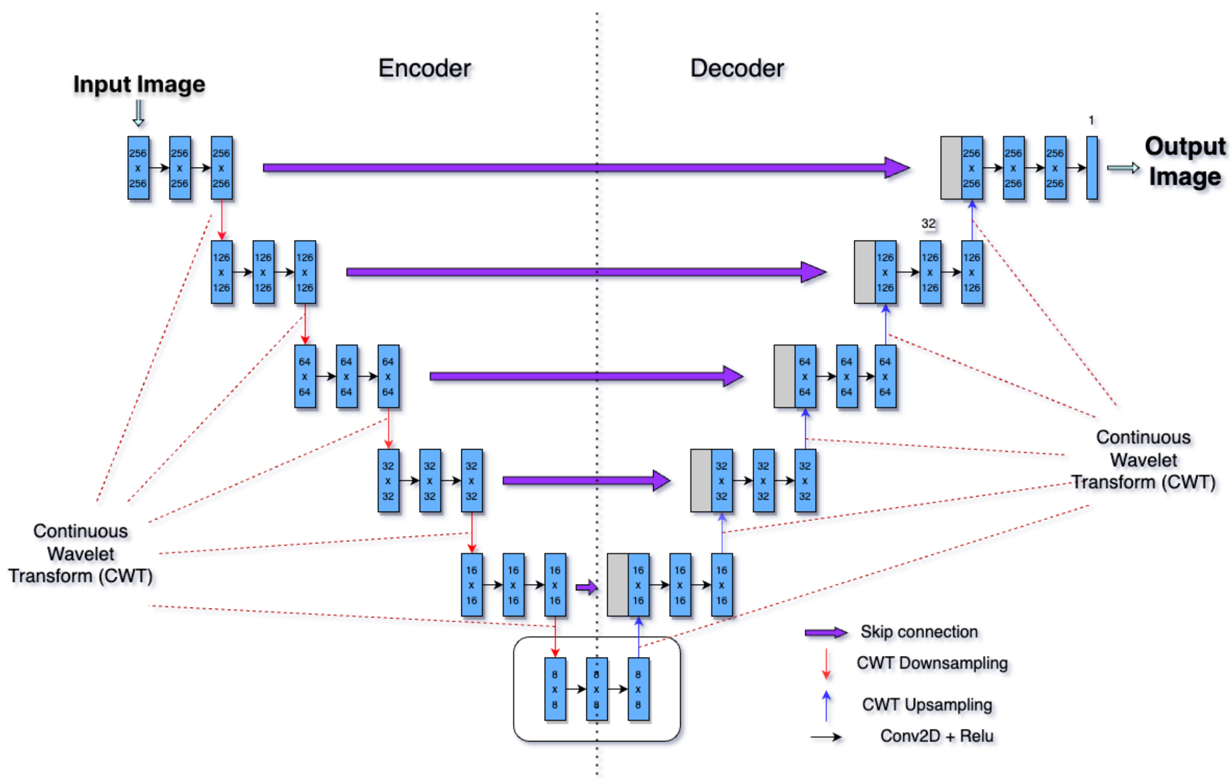

Fig. 7: Proposed Unet Architecture

The U-net architecture with skip layers is utilized to decompose maternal abdominal electrocardiograms (maECG) into fetal electrocardiograms (fECG). Figure 7 illustrates the suggested neural network architecture.

Using encoders, fECG features were obtained across multiple resolutions, and at each step, continuous wavelet transform was applied, performing Multi-Resolution Analysis (MRA) to capture signal characteristics at different resolutions. The signals were then reconstructed using decoders. Since each layer had a channel size of 16, 32, 64, 128, 256, and 512, features were extracted at various scales during the encoding phase.The channel size in the network used to extract the fECG was set to 1024 in the specific section with the highlighted black box (as depicted in Figure 7). More skip layers were added to improve performance, allowing the decoder to receive characteristics from the encoder at both high and low levels. For fECG extraction, each encoder's kernel size was set to 5, as a smaller kernel size can capture high resolution signal features, which are necessary given the small amplitude of fECG signals. In order to effectively separate the fECG, smaller kernel sizes were employed in the fECG extraction network. The signal amplitude was maintained with zero padding, and the stride was kept at one. Batch normalization was applied to every layer of convolution and deconvolution. The activation function of the leaky rectified linear unit was employed, with an alpha set at 0.02. The integration of continuous wavelet transform at each step in the encoder and decoder facilitates Multi-Resolution Analysis, reducing model complexity without compromising performance in both encoding and decoding phases.

F. Training Parameters

We chose an optimal batch size of 32 after taking into account a number of variables, including the size of the dataset, the available computing power, and the requirements of the U-Net model. The loss function was calculated using mean absolute error for optimizing the suggested network. It is less vulnerable to outliers and assigns equal weight to all errors. MAE is advantageous in minimizing the total error in the model while avoiding significant mistakes which outperforms mean squared error [19]. We used Adam optimizer and our training parameters were set at 0.001 for learning rate and a batch size of 32.

IV. Results

In Fig. 8 shows the Abdominal ECG, ground truth fetal ECG and predicted fetal ECG where the sample size of 3072. There have been numerous approaches where scalogram or spectrogram of ECG signals were used as an input and various architectures like CNN that have been implemented are capable of finding the fetal QRS peaks but such novel models are unable to predict P and T waves due to such high complexities. The proposed model was able to identify the fetal QRS peaks (highlighted by the black outlined box in Fig. 8). P waves, having a very small amplitude, is a lot difficult to find. As highlighted by the red circle in Fig. 8, it

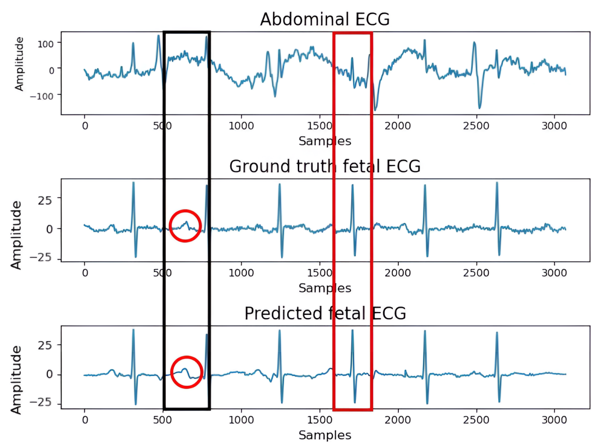

Fig. 8: Comparison of Abdominal ECG, ground truth fetal ECG and predicted fetal ECG

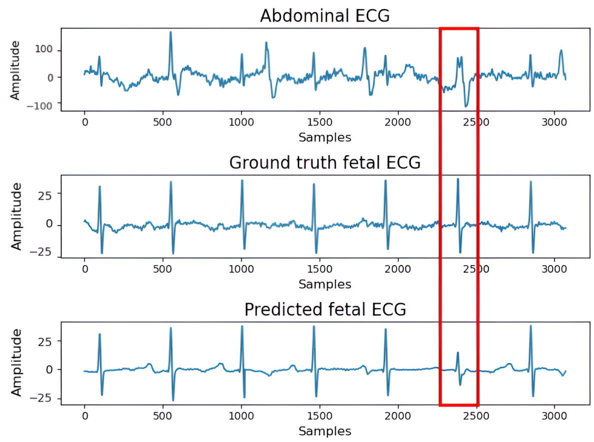

Fig. 9: Incorrect amplitude prediction of fetal ECG with ground truth fetal ECG

can be seen that the proposed model is also able to detect P waves. We obtained a recall, precision and F1 score of 91.88%, 94.36% and 93.1% respectively. There is still some room for improvisations needed as our proposed model sometimes wasn't able to predict the amplitude of QRS complexes of the waves correctly, which can be seen in the Fig. 9.

Also with the use of wavelets, the model's performance could potentially be enhanced further by incorporating more sophisticated wavelet transformations and exploring different types of wavelet families.

V. Conclusion

We put out a comprehensive neural network model designed to efficiently extract fECG signals from maECG data. Effective separation and extraction of fECG signals from actual and simulated datasets is possible. High F1 scores and accuracy values were also seen in the P and R-peak identification in fECG data. Therefore, the proposed deep learning architecture is a valuable tool for ongoing concurrent cardiac monitoring of the fetus. It is anticipated that this model will contribute to commercial applications when it is used in software that implements human-computer interaction. Finding a method for reducing the distortion in the fECG signal forms is one of the work's limitations, which is something we want to address in further study. Additionally, the project's future scope may involve usage of wavelet transformations to enhance model's performance.

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